

LƯƠNG QUỐC DŨNG

AI Engineer | Software Engineer

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OBJECTIVE

Seeking an AI Engineer internship where I can build production-ready AI applications using LLMs, RAG, AI Agents, and modern backend technologies. Interested in Generative AI, AI systems, and scalable software engineering.

EDUCATION

Thang Long University — B.Sc. in Information Technology 2021 – 2025

Focus: Information Systems, Data Analysis, Web Applications. Graduation project: *TripNest AI — AI Travel Agent*.

TECHNICAL SKILLS

AI Engineering: LLMs (Claude, OpenAI) • RAG (LangChain, ChromaDB) • AI Agents & Tool Calling • Prompt Engineering & Evaluation • Recommendation Systems

Backend: Python, FastAPI, REST APIs, SSE Streaming, SQLAlchemy, Node.js, Pytest

Frontend: React, Next.js 14, TypeScript, TailwindCSS, HTML/CSS/JavaScript

Data: PostgreSQL (Supabase), MySQL, ChromaDB, Pandas, NumPy, Scikit-learn

Cloud & DevOps: Docker, Fly.io, Vercel, GitHub Actions (CI), Git/GitHub

Languages: English — technical reading, documentation, and basic professional communication

EXPERIENCE

System Operations Collaborator — Viettel Customer Service 06/2025 – Present

- Operate and support V-Office, an enterprise document system for the Ministry of Justice and the Central Party Office.
- Manage user data and role-based permissions; resolve operational issues during digital rollout.

Front-end Developer Intern — OnTik 05/2024 – 12/2024

- Developed LoanCRM, an enterprise loan-management system; turned requirements into UI and flows.
- Integrated REST APIs with backend services and databases; collaborated effectively in an Agile team.

PROJECTS

TripNest AI — AI Travel Agent (Graduation Project)

Live: tripnesttravel.online • GitHub: github.com/quocdungTlu/C2-App-031

FastAPI • Next.js 14 • Claude Haiku 4.5 / GPT-4.1 mini • LangChain • ChromaDB • Supabase PostgreSQL • Docker • Fly.io • Vercel

- Built and deployed to production an AI travel agent: LLM tool calling, RAG, SSE streaming, and 11 agent tools covering room search, pricing, policy Q&A, itineraries, and a booking flow with VietQR payments and Google/Facebook SSO.
- Architected a multi-provider LLM layer behind a feature flag (Claude Haiku 4.5 / GPT-4.1 mini) with a canonical message format enabling safe A/B testing and rollback; prompt caching cut cached input costs ~90% (~\$0.17/user/month, documented cost report).
- Engineered LLM context management: history summarization, tool-result pruning, cache-friendly prompt structure, and DB-persisted sessions with TTL eviction.
- Built personalization: Guest DNA profiles with implicit preference learning, explainable Match Score recommendations, and multi-session memory; prevented overbooking atomically with PostgreSQL advisory locks.
- Instrumented telemetry and an admin analytics dashboard: per-turn token/cost/latency tracking, tool success rate, conversion funnel, and deterministic A/B prompt experiments.
- Hardened safety and anti-hallucination layers (pre-LLM credit-card blocking, prompt-injection defense): 100% retrieval accuracy on an internal 20-question evaluation set, 9/9 end-to-end booking flows, 255 automated tests (unit + integration).

E-commerce Website — Computer Components Store (ReactJS • Vite • github.com/quocdungTlu/CDTN)

- Built the storefront UI and optimized the product-display flow.

LEADERSHIP

Vice President — Karatedo Club, Thang Long University — lead club operations and organize events 2024 – Present